

AIR CONDITIONING SPECIFICATIONS

1. SCOPE OF WORK

THIS CONTRACTOR SHALL FURNISH ALL LABOR, MATERIALS, TOOLS, MACHINERY, EQUIPMENT, APPLIANCES AND SERVICES AS NECESSARY TO COMPLETE THE AIR CONDITION WORK UNDER THIS CONTRACT.

2. EXAMINATION OF SITE

BEFORE BIDDING ON THIS WORK, THIS CONTRACTOR SHALL THOROUGHLY FAMILIARIZE HIMSELF WITH THE REQUIREMENTS OF THE PROJECT. BY THE ACT OF SUBMITTING A PROPOSAL FOR THE WORK INCLUDED IN THIS CONTRACT, THE CONTRACTOR SHALL BE DEEMED TO HAVE MADE SUCH STUDY AND EXAMINATION, AND TO ACCEPT ALL CONDITIONS PRESENTED IN THE DRAWINGS, HEREIN SPECIFIED, OR AT THE SITE. NO REQUEST FOR ADDITIONAL PAYMENT WILL BE CONSIDERED AS VALID DUE TO FAILURE TO ALLOW FOR CONDITIONS WHICH MAY EXIST. ALL OPENINGS THROUGH ROOFS AND WALLS MUST BE APPROVED BY THE STRUCTURAL ENGINEER.

3. PERMITS AND INSPECTIONS

- A. ALL WORK MUST COMPLY WITH REQUIREMENTS OF ALL APPLICABLE CODES, LAWS, ORDINANCES, REGULATIONS AND OTHER AUTHORITIES HAVING JURISDICTION.
- B. THIS CONTRACTOR SHALL SECURE AND PAY FOR ALL REQUIRED LICENSES AND PERMITS INVOLVED WITH THE PERFORMANCE OF HIS WORK ON THIS PROJECT, AND SHALL ARRANGE FOR ALL INSPECTIONS AS REQUIRED BY THE ENFORCEMENT AGENCIES.
- C. CORRECTIONS FOR ITEMS OF WORKMANSHIP OR METHOD OF INSTALLATION THAT MAY BE REQUIRED BY THE LOCAL BUILDING AUTHORITIES SHALL BE PROMPTLY ACCOMPLISHED WITHOUT ADDITIONAL COST TO THE OWNER.

4. COORDINATION

THIS CONTRACTOR SHALL WORK CLOSELY WITH ALL TRADES, MAKING CERTAIN ALL ITEMS ARE COVERED. NO EXTRA WILL BE ALLOWED FOR ANY CONTROVERSIES ARISING BETWEEN THE TRADES.

5. CLOSING IN WORK

THIS CONTRACTOR SHALL NOT ALLOW ANY OF HIS WORK TO BE ENCLOSED OR COVERED UP UNTIL APPROVAL BY ENFORCING AGENCIES.

6. RESPONSIBILITY

THIS CONTRACTOR SHALL BE RESPONSIBLE FOR THE SAFETY AND GOOD CONDITION OF ALL MATERIAL AND EQUIPMENT OF THE ENTIRE INSTALLATION UNTIL FINAL ACCEPTANCE BY THE OWNER. HE SHALL ERECT AND MAINTAIN SUITABLE BARRIERS, PROTECTIVE DEVICES, LIGHTS, WARNING SIGNS WHERE REQUIRED FOR THE PROTECTION OF THE PUBLIC AND EMPLOYEES ABOUT THE BUILDING, AND SHALL BE FULLY RESPONSIBLE FOR ANY LOSS OR INJURY TO PERSONS OR PROPERTY RESULTING FROM HIS NEGLIGENCE OF THESE PRECAUTIONS HIS OWN CARELESSNESS, OR HIS EMPLOYEE'S CARELESSNESS OR NEGLIGENCE.

7. VERIFICATION OF DIMENSIONS

ALL SCALED AND FIGURED DIMENSIONS ARE APPROXIMATE AND ARE GIVEN FOR ESTIMATING PURPOSES ONLY. BEFORE PROCEEDING WITH THE WORK, THE CONTRACTOR SHALL CAREFULLY CHECK AND VERIFY ALL DIMENSIONS AND SIZES AND SHALL ASSUME ALL RESPONSIBILITIES

8. CUTTING AND PATCHING

- A. DO ALL CUTTING AND PATCHING REQUIRED FOR INSTALLATION OF WORK UNDER THIS CONTRACT. PATCHING IS TO MATCH SURROUNDING MATERIALS AND FINISH.

- B. HOLES OR NOTCHES SHALL NOT BE CUT IN STRUCTURAL MEMBERS WITHOUT WRITTEN CONSENT OF THE STRUCTURAL ENGINEER.

- C. CONTRACTOR SHALL ARRANGE FOR AND PLACE ALL NECESSARY SLEEVES OR OPENINGS IN THE STRUCTURE WHERE ALLOWED.

9. WORKMANSHIP AND MATERIALS

- A. ALL WORKMANSHIP SHALL BE FIRST CLASS IN ALL RESPECTS AND CARRIED OUT IN A MANNER SATISFACTORY TO AN MEETING THE APPROVAL OF THE ARCHITECT.
- B. ALL MATERIALS, APPLIANCES AND EQUIPMENT SHALL BE NEW AND THE BEST OF THEIR RESPECTIVE KIND, FREE FROM ALL DEFECTS AND OF THE MAKE, BRAND AND QUALITY SPECIFIED DIFFERENTLY AS ACCEPTED BY THE ARCHITECT AS PROVIDED.
- C. EXCEPT AS OTHERWISE SPECIFIED, ALL MATERIALS USED IN THE FABRICATION AND CONSTRUCTION OF THE VARIOUS PARTS OF THE EQUIPMENT INCLUDED IN THIS WORK SHALL BE IN ACCORDANCE WITH THE LATEST STANDARD SPECIFICATIONS OF THE AMERICAN SOCIETY FOR TESTING MATERIAL.

10. SHOP DRAWINGS

CONTRACTOR SHALL SUBMIT FIVE (5) COPIES OF A TYPEWRITTEN LIST COMPLETE FOR ALL MATERIALS AND EQUIPMENT SPECIFIED AND LISTED FOR WHICH HE PROPOSES TO INSTALL. THESE LISTS SHALL SET FORTH THE SPECIFICATION PAGE NUMBER; MANUFACTURER'S NAME; MODEL NUMBER; SIZES; NONSTANDARD ACCESSORIES SPECIFIED; AND SUCH OTHER INFORMATION REQUIRED TO IDENTIFY ALL ITEMS.

11. SHEET METAL WORK

SHEET METAL WORK SHALL BE GALVANIZED SHEETS OR FLEXIBLE DUCT, APPROVED BY THE ARCHITECT AND SHALL BE CONSTRUCTED AND INSTALLED IN ACCORDANCE WITH THE LATEST EDITION OF THE SMACNA DUCT MANUAL AND APPLICABLE CODES.

12. INSULATION

- A. ALL SUPPLY AND RETURN DUCTS INSTALLED ABOVE SUSPENDED CEILINGS AND IN FURRED SPACES SHALL BE COVERED ON ALL SIDES WITH 1-INCH THICK ULTRALITE NO. 150, 1-1/2 LB. PER CU. FT. DENSITY, FLEXIBLE DUCT INSULATION, OR EQUIVALENT FIBER GLASS, PITTSBURGH OR JOHNS-MANVILLE INSULATION SHALL BE WIRED ON 12-INCH CENTERS WITH NO. 16 GALVANIZED WIRE PRE-BENT TO TWO 90 DEGREE ANGLES OR CORNERS.
- B. DUCT LINING ONE INCH THICK, 1 1/2 LB DENSITY FIBER GLASS, MADE BY CSG ULTRALITE, KNUF TYPE M, JOHNS-MANVILLE OR EQUAL.

13. DAMPERS

- A. VOLUME DAMPERS OF THE SAME WEIGHT IRON AS THE DUCT SHALL BE INSTALLED WHERE SHOWN ON DRAWINGS AND /OR AS REQUIRED FOR BALANCING.
- B. DAMPER RODS SHALL BE 3/8" AND SHALL BE FITTED WITH VENT LOCK OR DURO-DYNE BEARING ON END FOR DAMPERS 14" LONG AND UNDER. FOR DAMPERS 15" LONG AND OVER, THE RODS SHALL BE FITTED WITH VENT-LOCK OR DURO-DYNE BEARINGS ON EACH SIDE OF THE DUCT.

14. GRILLES, REGISTERS AND DIFFUSERS

- A. THE CONTRACTOR SHALL FURNISH AND INSTALL ALL DIFFUSERS, REGISTERS AND GRILLES FOR THE AIR SUPPLY, AIR EXHAUST AND AIR RETURN INLETS AND OUTLETS OF THE TYPES, SIZES AND IN THE LOCATIONS INDICATED ON THE DRAWINGS.
- B. ALL REGISTERS, DIFFUSERS AND GRILLES SHALL BE INSTALLED IN APPROXIMATELY THE LOCATIONS INDICATED ON THE DRAWINGS BUT THE CONTRACTOR SHALL VERIFY EXACT LOCATIONS WITH THE ARCHITECT'S PLANS, MAKE ANY CHANGES AS MAY BE REQUIRED. DIFFUSER, REGISTER AND GRILLE FRAMES SHALL NOT OVERLAP THE CEILING TILE THE BAR SUPPORT BARS.
- C. OUTLETS (TO BE APPROVED BY ARCHITECT) TITUS, KRUEGER OR EQUAL APPROVED MANUFACTURERS.

15. EQUIPMENT

SEE SCHEDULE OF EQUIPMENT USED ON THE DRAWINGS.

16. CLEANING

- A. ALL NEW EQUIPMENT, DUCTWORK, ETC., SHALL BE THOROUGHLY CLEANED TO REMOVE RUST, SCALE, PLASTER OR ANY INTERNAL OBSTRUCTION BEFORE ANY COVERING IS INSTALLED OR ANY DUCTWORK OR EQUIPMENT IS PAINTED.
- B. ALL PARTS OF THE WORK WHICH ARE TO BE PAINTED OR WHICH ARE EXPOSED IN THE FINISHED WORK SHALL BE THOROUGHLY CLEANED AND MADE READY TO RECEIVE PAINT FINISH.

17. AIR BALANCE

A COMPLETE AIR BALANCE SHALL BE PERFORMED IN ACCORDANCE WITH THE ASSOCIATE AIR BALANCE COUNCIL NATIONAL STANDARD PROCEDURES AND SHALL BE MADE BY NATIONAL AIR BALANCE AND THE COST THERE OF INCLUDED IN THE CONTRACT.

18. OPERATING INSTRUCTIONS AND OPERATIONAL READINESS

- A. OPERATING INSTRUCTIONS
 - 1. AFTER THE ENTIRE SYSTEMS HAVE BEEN INSTALLED AND COMPLETED AND ALL CONNECTIONS MADE, THE SYSTEMS SHALL BE OPERATED CONTINUOUSLY FOR A PERIOD OF FOUR HOURS AT A TIME REQUESTED BY THE OWNER, AND FULFILL ALL GUARANTEE REQUIREMENTS.
 - 2. ALL POWER REQUIRED FOR THE ABOVE SPECIFIED OPERATING PERIOD WILL BE FURNISHED.
- B. OPERATIONAL READINESS
 - 1. THE CONTRACTOR SHALL INSURE THAT THE COMPLETE INSTALLATION, INCLUDING ALL EQUIPMENT AND CONTROLS IS COMPLETE, OPERATING, CHECKED AND ADJUSTED AT THE TIME OF FINAL INSPECTION.
 - 2. THE CONTRACTOR SHALL BE HELD RESPONSIBLE FOR ANY DELAYS INCURRED AND/OR RE-INSPECTIONS REQUIRED DUE TO ANY LACK OF ABOVE MENTIONED READINESS.

19. GUARANTEE

THIS CONTRACTOR SHALL UNDONCTIONALLY GUARANTEE ALL LABOR AND MATERIALS FURNISHED BY HIM AGAINST DEFECTS IN WORKMANSHIP OR DEFECTIVE MATERIALS FOR A PERIOD OF ONE YEAR FROM DATE OF FINAL INSPECTION APPROVAL.

20. AS-BUILT DRAWINGS

- A. THE AS-BUILT DRAWINGS SHALL BE PREPARED BY THE CONTRACTOR.
- B. THE CONTRACTOR SHALL INDICATE ON THE "AS-BUILT" DRAWINGS ALL DELETIONS IN RED. ALL ADDITIONS, RELOCATIONS, RE-ROUTING AND MODIFICATIONS SHALL BE INDICATED IN GREEN.

21. OPERATING-SERVICE-MAINTENANCE BROCHURES

- A. THE CONTRACTOR SHALL PROVIDE FIVE (5) COPIES OF THE OPERATING-SERVICE-MAINTENANCE BROCHURES FOR THE EQUIPMENT INSTALLED BY HIM. THE BROCHURES SHALL BE BOUND AND INDEXED AND PROVIDED WITH DIVIDER SHEETS WITH TABS.

Onuma & Associates

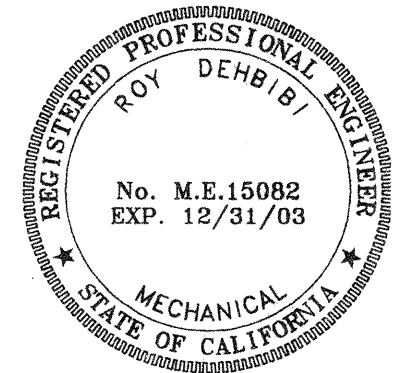
Architecture, Planning, Consulti
511 Mission Street
South Pasadena, CA 9103
tel 626 799 9920
fax 626 799 8260
e-mail: info@onuma.co

Consultants

MECHANICAL & ELECTRICAL ENGINEERS
DEBBI & ROSHANIAN INC.
North Hollywood, CA 9160
TEL (818) 980-0837
FAX (818) 508-7251

REMODEL
FOR
SAINT EDMUND'S
EPISCOPAL CHURCH
1175 SAN GABRIEL BLVD. SAN MARINO CA. 91108-8048

STAMPS



REVISIONS

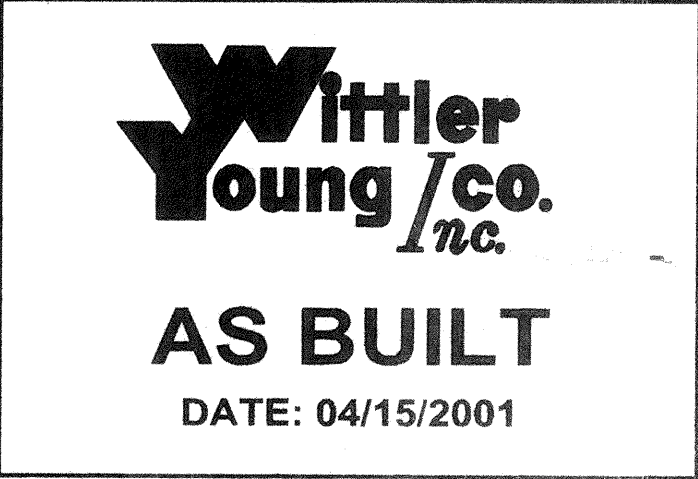
DESCRIPTION	DATE

PROJECT NUMBER 2000

SCALE AS NOTED

PRINTED 01/18/0

TITLE
MECHANICAL
SPECIFICATION
SHEET



M-0.1
OF